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Inventor(s)	Wu; Shang-Tsai

Laser marking feeding device

Abstract

A laser marking feeding device used for multiple parts to be marked includes a device body, a laser marking machine, a transmitting mechanism and a part blocking structure. The device body is disposed with a processing area and a feeding area. The transmitting mechanism is installed in the processing area and the feeding area and disposed under the laser marking machine. The transmitting mechanism is used for transmitting the parts to be marked from the feeding area to the processing area. The multiple parts to be marked are stacked on the transmitting mechanism of the feeding area. The part blocking structure is disposed between the processing area and the feeding area and disposed over the transmitting mechanism. An interval greater than a thickness of one part to be marked and less than thicknesses of two parts to be marked exists between the part blocking structure and the transmitting mechanism.

Inventors: Wu; Shang-Tsai (New Taipei, TW)

Applicant: DINKLE ENTERPRISE CO., LTD. (New Taipei, TW); DINKLE ELECTRIC MACHINERY (CHINA) CO., LTD. (Jiangsu, CN); LI YANG ELECTRIC MACHINERY (DONGGUAN) CO., LTD. (Dong Guan, CN)

Family ID: 1000007814312

Assignee: DINKLE ENTERPRISE CO., LTD. (New Taipei, TW); DINKLE ELECTRIC MACHINERY (CHINA) CO., LTD. (Jiangsu, CN); LI YANG ELECTRIC MACHINERY (DONGGUAN) CO., LTD. (Dong Guan, CN)

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Primary Examiner: Adams; Gregory W

Attorney, Agent or Firm: hdl services

Background/Summary

BACKGROUND

Technical Field

(1) The disclosure relates to a part feeding device of mechanical process, particularly to a laser marking feeding device.

Related Art

(2) Laser marking uses laser to form characters, patterns and other marks on surfaces of objects. The principle is to engrave a surface of an object with laser. Depending on the power and beam density of laser, different engraving texture effects can be produced.

(3) In addition, the laser marking process usually uses an automatic feeding device to feed the parts to be marked to the place under the laser marking machine one by one, and then uses the laser marking machine to engrave the surfaces of the parts to be marked. Finally, the parts which have been marked are removed from the part belt to finish the laser marking process.

(4) However, the above-mentioned automatic feeding device ensures that the parts to be marked are sent to the place under the laser marking machine one piece at a time, which often leads to problems such as the transmitting path of the automatic feeding device being too long or the device structure being complex. Therefore, how to simplify the transmitting path or structure of the automatic feeding device is the research and development focus of the automatic feeding device manufacturers.

(5) In view of this, the inventors have devoted themselves to the above-mentioned prior art, researched intensively and cooperated with the application of science to try to solve the above-mentioned problems. Finally, the invention which is reasonable and effective to overcome the above drawbacks is provided.

SUMMARY

(6) The disclosure provides a laser marking feeding device, which uses when the multiple parts to be marked are stacked on the transmitting mechanism in the feeding area, only the lowest one of the parts to be marked can be driven by the transmitting mechanism to move to pass through the interval, and the rest of the multiple parts to be marked are blocked in the feeding area by the part blocking structure to make the laser marking feeding device of the disclosure have advantages of simplifying the transmitting path of the automatically feeding or the device structure.

(7) In embodiments of the disclosure, the disclosure provides a laser marking feeding device used for multiple parts to be marked. The laser marking feeding device includes: a device body, disposed with a processing area and a feeding area disposed on a side of the processing area; a laser marking machine, installed in the processing area; a transmitting mechanism, installed in the processing area and the feeding area and disposed under the laser marking machine, used for transmitting the parts to be marked from the feeding area to the processing area along a horizontal path; and a part blocking structure, disposed between the processing area and the feeding area and disposed over the transmitting mechanism, a height of the part blocking structure being greater than thicknesses of the multiple parts to be marked, and an interval greater than a thickness of one of the parts to be marked and less than thicknesses of two of the parts to be marked existing between the part blocking structure and the transmitting mechanism; wherein when the multiple parts to be marked are stacked on the transmitting mechanism in the feeding area, only the lowest one of the parts to be marked can be driven by the transmitting mechanism to move from the feeding area to the processing area via the interval along the horizontal path, and the rest of the multiple parts to be marked are blocked in the feeding area by the part blocking structure.

(8) Accordingly, the laser marking feeding device of the disclosure adopts the transmitting mechanism to transmit the parts to be marked from the feeding area to the processing area along the horizontal path, and then uses that the height of the part blocking structure is greater than the thicknesses of the multiple parts to be marked and that an interval greater than a thickness of one part to be marked and less than thicknesses of two parts to be marked exists between the part blocking structure and the transmitting mechanism, after that, when the multiple parts to be marked are stacked on the transmitting mechanism in the feeding area, only the lowest one of the parts to be marked can be driven by the transmitting mechanism to move from the feeding area to the processing area via the interval along the horizontal path, and the rest of the multiple parts to be marked are blocked in the feeding area by the part blocking structure, so as to accomplish the function of automatically sending the parts to be marked into the processing area one by one. As a result, the laser marking feeding device of the disclosure has an advantage of simplifying the transmitting path of automatically feeding or the device structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective schematic view of the laser marking feeding device of the disclosure;
- (2) FIG. 2 is another perspective schematic view of the laser marking feeding device of the disclosure;
- (3) FIG. 3 is a perspective schematic view of the first (second) elevated blocking sheet module of the disclosure;
- (4) FIG. 4 is a schematic view of using status of the first (second) elevated blocking sheet module of the disclosure;
- (5) FIG. 5 is a cross-sectional view of the first using status of the laser marking feeding device of the disclosure;
- (6) FIG. 6 is a partially enlarged schematic view of FIG. 4 of the disclosure;
- (7) FIG. 7 is another schematic view of using status of the first (second) elevated blocking sheet module of the disclosure;
- (8) FIG. 8 is a cross-sectional view of the second using status of the laser marking feeding device of the disclosure;
- (9) FIG. 9 is a cross-sectional view of the third using status of the laser marking feeding device of the disclosure; and
- (10) FIG. 10 is a cross-sectional view of the fourth using status of the laser marking feeding device of the disclosure.

DETAILED DESCRIPTION

- (11) The technical contents of this disclosure will become apparent with the detailed description of embodiments accompanied with the illustration of related drawings as follows. It is intended that the embodiments and drawings disclosed herein are to be considered illustrative rather than restrictive.
- (12) Please refer to FIGS. 1-10. The disclosure provides a laser marking feeding device 10 includes a device body 1, a laser marking machine 2, a transmitting mechanism 3 and a part blocking structure 4.
- (13) As shown in FIGS. 1-2, 5-6 and 8-10, the laser marking feeding device 10 is used for multiple parts to be marked 100. Each part to be marked 100 has two transverse part belts 101 formed on two opposite sides thereof, multiple longitudinal part belts 102 connected between the two transverse part belts 101 and multiple materials to be laser marked 103 connected between the longitudinal part belts 102.
- (14) As shown in FIGS. 1-2, 5-6 and 8-10, the device body 1 is disposed with a processing area 11 and a feeding area 12 disposed on a side of the processing area 11.
- (15) As shown in FIGS. 2, 5 and 9, the laser marking machine 2 is installed in the processing area 11. The laser marking machine 2 is used for laser engraving surfaces of the multiple materials to be laser marked 103.
- (16) As shown in FIGS. 2, 5-6 and 8-10, the transmitting mechanism 3 is installed in the processing area 11 and the feeding area 12 and disposed under the laser marking machine 2. The transmitting mechanism 3 is used for transmitting the parts to be marked 100 from the feeding area 12 to the processing area 11 along a horizontal path.
- (17) In detail, the transmitting mechanism 3 includes two driving wheels 31 and two driven wheels 32, which are pivoted to the device body 1 and disposed on two opposite sides of the processing area 11 and the feeding area 12, and two tracks 33 separately wrapping outsides of each driving wheel 31 and each driven wheel 32 and following each driving wheel 31 to rotate. One of the driving wheels 31, one of the driven wheels 32 and one of the tracks 33 are disposed on one side of the processing area 11 and the feeding area 12, and the other one of the driving wheels 31, the other one of the driven wheels 32 and the other one of the tracks 33 are disposed on the other side of the processing area 11 and the feeding area 12. The two transverse part belts 101 are placed on the two

tracks **33** to make the transmitting mechanism **3** able to transmit the parts to be marked **100** from the feeding area **12** to the processing area **11** along the horizontal path. The track **33** of the embodiment may be a belt or a chain.

(18) As shown in FIGS. **1-2**, **5-6** and **8-10**, the part blocking structure **4** is disposed between the processing area **11** and the feeding area **12** and disposed over the transmitting mechanism **3**. A height **H** of the part blocking structure **4** is greater than the sum of thicknesses of the multiple parts to be marked **100**. An interval **S** greater than a thickness of one of the parts to be marked **100** and less than thicknesses of two of the parts to be marked **100** exists between the part blocking structure **4** and the transmitting mechanism **3**.

(19) Further, the part blocking structure **4** includes two upright blocking plates **41** fixed to the device body **1** and disposed over the two tracks **33**. The height **H** is a height of each upright blocking plate **41**. The interval **S** is formed between the two tracks **33** and the two upright blocking plates **41**.

(20) When the multiple parts to be marked **100** are stacked on the transmitting mechanism **3** of the feeding area **12**, only the lowest one of the parts to be marked **100** can be driven by the transmitting mechanism **3** to move from the feeding area **12** to the processing area **11** via the interval **S** along the horizontal path. The rest of the multiple parts to be marked **100** are blocked in the feeding area **12** by the part blocking structure **4**.

(21) As shown in FIGS. **1-2**, **5** and **8-10**, the laser marking feeding device **10** further includes three limiting plates **5** fixed to the device body **1**. One of the limiting plates **5** and the two upright blocking plates **41** are disposed on a pair of two opposite sides of the feeding area **12** and its distal end is extended with a slant section **51** inclining upward gradually outward to be advantageous to stacking the multiple parts to be marked **100** on the transmitting mechanism **3** in the feeding area **12**. The remaining two limiting plates **5** are disposed on another pair of two opposite sides of the feeding area **12**.

(22) As shown in FIGS. **2-10**, the laser marking feeding device **10** further includes a first elevated blocking sheet module **6** disposed among the two tracks **33**, the processing area **11** and the feeding area **12**. The first elevated blocking sheet module **6** includes a first blocking sheet **61** and a first actuator **62** fixed to the device body **1** and driving the first blocking sheet **61** to be higher than or lower than the two tracks **33**. The first blocking sheet **61** and the two upright blocking plates **41** are located on the same horizontal plane or disposed on a side of the two upright blocking plates **41**, which is adjacent to the feeding area **12**.

(23) In addition, the first elevated blocking sheet module **6** further includes a first position sensor **63** and one or more second position sensors **64**. The first position sensor **63** is installed on one side of the first actuator **62**, which is adjacent to the feeding area **12**. The first position sensor **63** is used for sensing the parts to be marked **100**. The second position sensor **64** corresponds to the first blocking sheet **61** to be disposed and is used for sensing the first blocking sheet **61** being higher than or lower than the two tracks **33**.

(24) In detail, the second position sensor **64** of the embodiment is two in number. The first blocking sheet **61** is disposed with a first notch **611**. One of the second position sensors **64** corresponds to the bottom edge of the first blocking sheet **61** to be disposed, and the other second position sensor **64** corresponds to the first notch **611** of the first blocking sheet **61** to be disposed. The second position sensors **64** may be infrared sensors or ultrasonic sensors. As shown in FIG. **4**, when the first blocking sheet **61** is higher than the two tracks **33**, the first blocking sheet **61** is higher than one of the second position sensors **64** to make the infrared rays of one of the infrared sensors or the ultrasonic waves of one of the ultrasonic sensors fail to be blocked to generate a first protrusion signal. The first blocking sheet **61** shades the other second position sensor **64** to make the infrared rays of the other infrared sensor or the ultrasonic waves of the other ultrasonic sensor blocked to generate a second protrusion signal. A processor (not shown in the figures) presets that the first blocking sheet **61** must be higher than the two tracks **33** by calculation and does not issue a

warning when simultaneously receiving the first and second protrusion signals. Contrarily, the processor presets that the first blocking sheet **61** must be higher than the two tracks **33** by calculation and issues a warning by a buzzer (not shown in the figures) or a display (not shown in the figures) when simultaneously not receiving the first and second protrusion signals.

(25) Furthermore, as shown in FIG. 7, when the first blocking sheet **61** is lower than the two tracks **33**, the first blocking sheet **61** shades one of the second position sensors **64** to make the infrared rays of one of the infrared sensors or the ultrasonic waves of one of the ultrasonic sensors blocked to generate a first retraction signal. The other second position sensor **64** corresponds to the first notch **611** of the first blocking sheet **61** to make the infrared rays of one of the infrared sensors or the ultrasonic waves of one of the ultrasonic sensors fail to be blocked to generate a second retraction signal. The processor (not shown in the figures) presets that the first blocking sheet **61** must be lower than the two tracks **33** by calculation and does not issue a warning when simultaneously receiving the first and second retraction signals. Contrarily, the processor presets that the first blocking sheet **61** must be lower than the two tracks **33** by calculation and issues a warning by a buzzer (not shown in the figures) or a display (not shown in the figures) when simultaneously not receiving the first and second protrusion signals.

(26) As shown in FIGS. 2-5 and 7-10, the laser marking feeding device **10** further includes a second elevated blocking sheet module **7** disposed between the two tracks **33** and on a side of the processing area **11**, which is away from the first elevated blocking sheet module **6**. The second elevated blocking sheet module **7** includes a second blocking sheet **71** and a second actuator **72** fixed to the device body **1** and driving the second blocking sheet **71** to be higher than or lower than the two tracks **33**.

(27) In addition, the second elevated blocking sheet module **7** further includes a third position sensor **73** and a fourth position sensor **74**. The third position sensor **73** is installed on a side of the second actuator **72**, which is adjacent to the processing area **11**. The third position sensor **73** is used for sensing the parts to be marked **100**. The fourth position sensor **74** corresponds to the second blocking sheet **71** to be disposed and is used for sensing the second blocking sheet **71** being higher than or lower than the two tracks **33**.

(28) In detail, the fourth position sensor **74** of the embodiment is two in number. The second blocking sheet **71** is disposed with a second notch **711**. One of the fourth position sensors **74** corresponds to the bottom edge of the second blocking sheet **71** to be disposed, and the other fourth position sensor **74** corresponds to the second notch **711** of the second blocking sheet **71** to be disposed. The fourth position sensors **74** may be infrared sensors or ultrasonic sensors. As shown in FIG. 4, when the second blocking sheet **71** is higher than the two tracks **33**, the second blocking sheet **71** is higher than one of the fourth position sensors **74** to make the infrared rays of one of the infrared sensors or the ultrasonic waves of one of the ultrasonic sensors fail to be blocked to generate a third protrusion signal. The second blocking sheet **71** shades the other fourth position sensor **74** to make the infrared rays of the other infrared sensor or the ultrasonic waves of the other ultrasonic sensor blocked to generate a fourth protrusion signal. The processor presets that the second blocking sheet **71** must be higher than the two tracks **33** by calculation and does not issue a warning when simultaneously receiving the third and fourth protrusion signals. Contrarily, the processor presets that the second blocking sheet **71** must be higher than the two tracks **33** by calculation and issues a warning by a buzzer or a display when simultaneously not receiving the third and fourth protrusion signals.

(29) Furthermore, as shown in FIG. 7, when the second blocking sheet **71** is lower than the two tracks **33**, the second blocking sheet **71** shades one of the fourth position sensors **74** to make the infrared rays of one of the infrared sensors or the ultrasonic waves of one of the ultrasonic sensors blocked to generate a third retraction signal. The other fourth position sensor **74** corresponds to the second notch **711** of the second blocking sheet **71** to make the infrared rays of the other infrared sensor or the ultrasonic waves of the other ultrasonic sensor fail to be blocked to generate a fourth

retraction signal. The processor presets that the second blocking sheet **71** must be lower than the two tracks **33** by calculation and does not issue a warning when simultaneously receiving the third and fourth retraction signals. Contrarily, the processor presets that the second blocking sheet **71** must be lower than the two tracks **33** by calculation and issues a warning by a buzzer (not shown in the figures) or a display (not shown in the figures) when simultaneously not receiving the third and fourth protrusion signals.

(30) Also, the first position sensor **63** will generate a first positioning signal when sensing the parts to be marked **100**. The third position sensor **73** will generate a non-positioning signal when not sensing the parts to be marked **100**. The fourth position sensor **74** will generate a descent signal when sensing the second blocking sheet **71** being lower than the two tracks **33**. The first actuator **62** will drive the first blocking sheet **61** to be lower than the two tracks **33** when receiving the first position signal and the non-positioning signal or receiving the first positioning signal and the descent signal. Except the abovementioned two situations, the first actuator **62** only drives the first blocking sheet **61** to be higher than the two tracks **33**.

(31) In addition, the third position sensor **73** will generate a second positioning signal when sensing the parts to be marked **100**. The first actuator **62** will drive the first blocking sheet **61** to be higher than the two tracks **33** when receiving the second positioning signal. The laser marking machine **2** will laser engrave the parts to be marked **100** when receiving the second positioning signal. The laser marking machine **2** will generate a laser finishing signal when finishing the laser engraving. The second actuator **72** will drive the second blocking sheet **71** to be lower than the two tracks **33** when receiving the second positioning signal and the laser finishing signal. Except the abovementioned situations, the second actuator **72** only drives the second blocking sheet **71** to be higher than the two tracks **33**.

(32) As shown in FIGS. 5-6 and 8, which show the first and second using statuses of the laser marking feeding device **10** of the disclosure, first, the multiple parts to be marked **100** are stacked on the transmitting mechanism **3** located in the feeding area **12**. The first position sensor **63** will generate the first positioning signal when sensing the parts to be marked **100**. At this time, there is no part to be marked **100** in the processing area **11**, so the third position sensor **73** fails to sense the parts to be marked **100** to generate the non-positioning signal. The first actuator **62** will drive the first blocking sheet **61** to be lower than the two tracks **33** when receiving the first positioning signal and the non-positioning signal. Then the two tracks **33** follow each driving wheel **31** to rotate, the lowest part to be marked **100** can be driven by the two tracks **33** to move from the feeding area **12** to the processing area **11** via the interval **S** along the horizontal path. The rest of the multiple parts to be marked **100** are blocked in the feeding area **12** by the part blocking structure **4**.

(33) As shown in FIGS. 9-10, which show the third and fourth using statuses of the laser marking feeding device **10** of the disclosure, finally, after the parts to be marked **100** enter the processing area **11**, the third position sensor **73** senses that the parts to be marked **100** will generate the second positioning signal, and the first actuator **62** will drive the first blocking sheet **61** to be higher than the two tracks **33** when receiving the second positioning signal to prevent another part to be marked **100** from being sent to the processing area **11**. The laser marking machine **2** will laser engrave the parts to be marked **100** when receiving the second positioning signal. The laser marking machine **2** will generate a laser finishing signal when finishing the laser engraving. The second actuator **72** will drive the second blocking sheet **71** to be lower than the two tracks **33** when receiving the second positioning signal and the laser finishing signal to make the finished parts to be marked **100** sent to a side of the processing area **11**, which is away from the feeding area **12**, so as to escape from the two tracks **33** to a collecting box (not shown in the figures).

(34) In addition, the fourth position sensor **74** will generate the descent signal when sensing the second blocking sheet **71** being lower than the two tracks **33**. The first actuator **62** will drive the first blocking sheet **61** to be lower than the two tracks **33** when receiving the first position signal and the descent signal to make another the lowest part to be marked **100** able to be driven by the

two tracks **33** to move from the feeding area **12** to the processing area **11** via the interval **S** along the horizontal path. The rest of the multiple parts to be marked **100** are blocked in the feeding area **12** by the part blocking structure **4**. When the finished part to be marked **100** are sent out of the processing area **11**, another part to be marked **100** can be sent into the processing area **11** to accomplish the function of process automation.

(35) Thereby, the laser marking feeding device **10** of the disclosure adopts the transmitting mechanism **3** to transmit the parts to be marked **100** from the feeding area **12** to the processing area **11** along the horizontal path, and then uses that the height **H** of the part blocking structure **4** is greater than the thicknesses of the multiple parts to be marked **100** and that an interval **S** greater than a thickness of one part to be marked **100** and less than thicknesses of two parts to be marked **100** exists between the part blocking structure **4** and the transmitting mechanism **3**, after that, when the multiple parts to be marked **100** are stacked on the transmitting mechanism **3** in the feeding area **12**, only the lowest one of the parts to be marked **100** can be driven by the transmitting mechanism **3** to move from the feeding area **12** to the processing area **11** via the interval **S** along the horizontal path, and the rest of the multiple parts to be marked **100** are blocked in the feeding area **12** by the part blocking structure **4**, so as to accomplish the function of automatically sending the parts to be marked **100** into the processing area **11** one by one. As a result, the laser marking feeding device **10** of the disclosure has an advantage of simplifying the transmitting path of automatically feeding or the device structure.

(36) While this disclosure has been described by means of specific embodiments, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope and spirit of this disclosure set forth in the claims.

Claims

1. A laser marking feeding device, used for multiple parts to be marked, the laser marking feeding device comprising: a device body, disposed with a processing area and a feeding area disposed on a side of the processing area; a laser marking machine, installed in the processing area; a transmitting mechanism, installed in the processing area and the feeding area and disposed under the laser marking machine, used for transmitting the parts to be marked from the feeding area to the processing area along a horizontal path, wherein the transmitting mechanism comprises two driving wheels and two driven wheels, which are pivoted to the device body and disposed on two opposite sides of the processing area and the feeding area, and two tracks separately wrapping outsides of each driving wheel and each driven wheel and following each driving wheel to rotate; a part blocking structure, disposed between the processing area and the feeding area and disposed over the transmitting mechanism, a height of the part blocking structure being greater than thicknesses of the multiple parts to be marked, and an interval greater than a thickness of one of the parts to be marked and less than thicknesses of two of the parts to be marked existing between the part blocking structure and the transmitting mechanism; and a first elevated blocking sheet module disposed among the two tracks, the processing area and the feeding area; wherein when the multiple parts to be marked are stacked on the transmitting mechanism in the feeding area, only the lowest one of the parts to be marked can be driven by the transmitting mechanism to move from the feeding area to the processing area via the interval along the horizontal path, and the rest of the multiple parts to be marked are blocked in the feeding area by the part blocking structure; wherein the part to be marked comprises two transverse part belts formed on two opposite sides thereof, and the two transverse part belts are placed on the two tracks; wherein the part blocking structure comprises two upright blocking plates fixed to the device body and disposed over the two tracks, the height is a height of each upright blocking plate, and the interval is formed between the two tracks and the two upright blocking plates; wherein the first elevated blocking sheet module comprises a first blocking sheet and a first actuator fixed to the device body and driving the first

blocking sheet to be higher than or lower than the two tracks, and the first blocking sheet and the two upright blocking plates are located on the same horizontal plane or disposed on a side of the two upright blocking plates, which is adjacent to the feeding area.

2. The laser marking feeding device of claim 1, further comprising three limiting plates fixed to the device body, wherein one of the limiting plates and the two upright blocking plates are disposed on a pair of two opposite sides of the feeding area and its distal end is extended with a slant section inclining upward gradually outward, and remaining two limiting plates are disposed on another pair of two opposite sides of the feeding area.

3. The laser marking feeding device of claim 1, wherein the first elevated blocking sheet module further comprises a first position sensor, the first position sensor is installed on one side of the first actuator, which is adjacent to the feeding area, and the first position sensor is used for sensing the parts to be marked.

4. The laser marking feeding device of claim 1, wherein the first elevated blocking sheet module further comprises at least one second position sensors, the second position sensor corresponds to the first blocking sheet to be disposed and is used for sensing the first blocking sheet being higher than or lower than the two tracks.

5. The laser marking feeding device of claim 3, further comprising a second elevated blocking sheet module disposed between the two tracks and on a side of the processing area, which is away from the first elevated blocking sheet module, wherein the second elevated blocking sheet module comprises a second blocking sheet and a second actuator fixed to the device body and driving the second blocking sheet to be higher than or lower than the two tracks.

6. The laser marking feeding device of claim 5, wherein the second elevated blocking sheet module further comprises a third position sensor, the third position sensor is installed on a side of the second actuator, which is adjacent to the processing area, and the third position sensor is used for sensing the parts to be marked.

7. The laser marking feeding device of claim 6, wherein the second elevated blocking sheet module further comprises a fourth position sensor, and the fourth position sensor corresponds to the second blocking sheet to be disposed and is used for sensing the second blocking sheet being higher than or lower than the two tracks.

8. The laser marking feeding device of claim 7, wherein the first position sensor generates a first positioning signal when sensing the parts to be marked, the third position sensor generates a non-positioning signal when not sensing the parts to be marked, the fourth position sensor generates a descent signal when sensing the second blocking sheet being lower than the two tracks, and the first actuator drives the first blocking sheet to be lower than the two tracks when receiving the first position signal and the non-positioning signal or receiving the first positioning signal and the descent signal.

9. The laser marking feeding device of claim 8, wherein the third position sensor generates a second positioning signal when sensing the parts to be marked, the first actuator drives the first blocking sheet to be higher than the two tracks when receiving the second positioning signal, the laser marking machine laser engraves the parts to be marked when receiving the second positioning signal, the laser marking machine generates a laser finishing signal when finishing the laser engraving, and the second actuator drives the second blocking sheet to be lower than the two tracks when receiving the second positioning signal and the laser finishing signal.
